Kernal Examples

① $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ $z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$

Feature map $\phi(x) = \begin{bmatrix} x_1x_1 \\ x_1x_2 \\ x_1x_3 \\ x_2x_1 \\ x_2x_2 \\ x_1x_3 \\ x_3x_1 \\ x_3x_2 \\ x_3x_3 \end{bmatrix}$ $\phi(z) = \begin{bmatrix} z_1z_1 \\ z_1z_2 \\ z_1z_3 \\ z_2z_1 \\ z_2z_2 \\ z_3z_3 \\ z_3z_1 \\ z_3z_2 \\ z_3z_3 \end{bmatrix}$

Solve for $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ $z = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$

$\phi(x) = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \\ 4 \\ 6 \\ 3 \\ 6 \\ 9 \end{bmatrix}$ $\phi(z) = \begin{bmatrix} 16 \\ 20 \\ 24 \\ 20 \\ 25 \\ 30 \\ 24 \\ 30 \\ 36 \end{bmatrix}$

$\phi(x) \cdot \phi(z) = 16 + 40 + 72 + 40 + 100 + 180 + 72 + 180 + 324 = 1024$

Kernal trick:

$k(x,z) = (\langle x, z \rangle)^2$

$k(x,z) = (4 + 10 + 18)^2$

$= 1024$

② $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$

$$\phi(x) = \begin{bmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2}\, x_1 x_2 \\ \sqrt{2}\, x_1 \\ \sqrt{2}\, x_2 \\ 1 \end{bmatrix}$$

for $x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $z = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

$$\phi(x) = \begin{bmatrix} 1 \\ 4 \\ 2\sqrt{2} \\ \sqrt{2} \\ 2\sqrt{2} \\ 1 \end{bmatrix} \qquad \phi(z) = \begin{bmatrix} 9 \\ 16 \\ 12\sqrt{2} \\ 3\sqrt{2} \\ 4\sqrt{2} \\ 1 \end{bmatrix}$$

$$\phi(x)^T \phi(z) = 9 + 64 + 48 + 6 + 16 + 1$$

$$= 144$$

$$K(x, z) = (x \cdot z + 1)^2$$

$$= ((3 + 8) + 1)^2$$

$$= (12)^2$$

$$= 144$$